

End-to-End Autonomous Job Discovery, Multi-Tier Scoring, Dynamic ATS Tailoring & Multi-Mode Applications

1. Master Workflow & Pipeline Architecture

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██████████ AI JOB AGENT ██████████
██████████ ▼
██████████ SEARCH MANAGER ██████████ (Reads
config/job_sources.yml Registry) ██████████
██████████ GLOBAL REMOTE JOBS ██████████
██████████ INDIAN JOBS ██████████ RECRUITER POSTS (HR) ██████████ (Jobicy, Remoteive, ██████████ (Adzuna, Naukri, ██████████ (LinkedIn Posts, ██████████
YC, Wellfound, ATS) ██████████ Foundit, Shine) ██████████ HackerNews Threads) ██████████
██████████ ▼
██████████ JOB NORMALIZER & DEDUPLICATOR ██████████ (Builds
Unified JobData & SQLite SHA-256 Check) ██████████ ▼
██████████ TODAY & FRESHNESS FILTER (0-24 Hrs) ██████████
██████████ ▼
██████████ EXPERIENCE ANALYZER (Strict 0-2 Years
ONLY) ██████████ (Hard-Blocks >=3 Yrs & Senior/Lead/Mgr Titles) ██████████
██████████ ▼
██████████ 2-TIER SCORER: ROLE & SKILL AI MATCHER ██████████
(30% Role + 25% Exp + 25% Skill + 20% Fit) ██████████
▼ ██████████ QUALIFIED MATCHES (Score >= 65) ██████████
██████████
██████████ NORMAL JOB POST ██████████ RECRUITER POST (EMAIL) ██████████
██████████ DYNAMIC ATS
RESUME PDF TAILORING ██████████
██████████ SOURCE-AWARE APPLICATION ENGINE ██████████
██████████
██████████ MODE 1: ██████████ MODE 2: ██████████ ☑️ MODE 3: ██████████ DIRECT ATS API ██████████ PLAYWRIGHT ██████████
██████████ VERIFIED RECRUITER ██████████ (Greenhouse/Lever ██████████ PERSISTENT ██████████ EMAIL OUTREACH ██████████ /Ashby Submit) ██████████ BROWSER
AUTO-FILL ██████████ (Gmail API OAuth) ██████████
██████████ ▼
MULTI-CHANNEL TELEMETRY ██████████ • WhatsApp Daily Report & Top Match Alerts ██████████ • Multi-Tab Excel Workbook
(openpyxl) ██████████ • Email Application Tracking Log ██████████

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2. Dynamic Source Configuration (job_sources.yml)

Instead of hardcoding source logic in Python, the agent loads app/config/job_sources.yml at runtime. Sources can be toggled on/off and prioritized dynamically without code modifications:

```
# app/config/job_sources.yml
sources:
  greenhouse: { enabled: true, type: career_page, priority: high }
  lever: { enabled: true, type: career_page, priority: high }
  ashby: { enabled: true, type: career_page, priority: high }
  workday: { enabled: true, type: career_page, priority: high }
  smartrecruiters: { enabled: true, type: career_page, priority: high }
  jobicy: { enabled: true, type: remote, priority: high }
  remotive: { enabled: true, type: remote, priority: high }
  himalayas: { enabled: true, type: remote, priority: high }
  adzuna: { enabled: true, type: job_board, priority: high }
  agent_reach: { enabled: true, type: recruiter_post, priority: high }
  linkedin_posts: { enabled: true, type: recruiter_post, priority: high }
  ycombinator: { enabled: true, type: remote, priority: high }
  wellfound: { enabled: true, type: remote, priority: high }
  arbeitnow: { enabled: true, type: remote, priority: medium }
  weworkremotely: { enabled: true, type: remote, priority: medium }
  remoteok: { enabled: true, type: remote, priority: medium }
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3. ExperienceAnalyzer & Weighted Multi-Dimensional Scoring

Dedicated ExperienceAnalyzer (Strict 0-2 Years Target):

The ExperienceAnalyzer inspects both the job title and full description. Any role requiring 3+ years, 4+ years, or ranges like 3–5 yrs / 2–4 yrs is rejected immediately before resume tailoring or application.

Requirement Pattern	Extracted Min / Max	Pipeline Action	Rationale & Behavior
Fresher / Entry / 0-2 yrs	Min: 0.0, Max: 2.0	PASS (Eligible)	Perfect target match. Advances to AI semantic scoring.
1-2 years / 2 years	Min: 1.0, Max: 2.0	PASS (Eligible)	Within candidate's 0-2 years background.
2-4 years / 2-5 years	Min: 2.0, Max: 4.0+	HARD REJECT	Exceeds 2-year upper threshold.
3+ years / 3-5 years	Min: 3.0, Max: 5.0	HARD REJECT	Experienced tier. Hard-blocked before LLM.
Senior / Staff / Lead / Mgr	Seniority Keyword	HARD REJECT	Senior title guard triggered. Score set to 0.

Multi-Dimensional Scoring Formula:

$$\text{Overall Score} = (30\% \times \text{Role Match}) + (25\% \times \text{Experience Match}) + (25\% \times \text{Skill Match}) + (10\% \times \text{Location Match}) + (10\% \times \text{Freshness Quality})$$

4. Source-Aware Application Engine & Dynamic ATS Resumes

100% Unique Dynamic ATS Resumes:

For every qualifying job (Score ≥ 65), Gemini extracts key technologies and tailors the summary, skill badges, and work accomplishments. ReportLab Platypus builds a pixel-perfect, 1-page ATS compliant PDF saved to resumes/.

Application Mode	Target Portals	Execution Strategy & Safety Rules
■ Mode 1: Direct API	Greenhouse, Lever, Ashby	Programmatic multipart form submission via official job board APIs with tailored ATS resume attached.
■ Mode 2: Persistent Browser	Wellfound, YC, LinkedIn, Workday, Jobicy	Playwright Chromium reuses authenticated session profile (data/browser_profile). Auto-fills forms and uploads tailored resume without password leakage.
✉■ Mode 3: Verified HR Email	LinkedIn Posts, HackerNews Hiring	Zero Bounce Policy: Sends cold email outreach ONLY to explicit recruiter emails extracted from post text. Never guesses synthetic addresses.
■ Mode 4: 1-Click Link Prep	Adzuna, Hirst, Shine, Cutshort, Foundit	Prepares custom ATS PDF resume + cover note and formats 1-click direct link in Excel report for anti-bot portals.

5. Production Codebase Directory Structure

Complete clean directory layout of the repository:

```

AI-Job-Agent/ ■■■■ app/ ■ ■■■■ agents/ # LangGraph Workflow Orchestrators ■ ■ ■■■■ nodes.py # 6 Pipeline
Nodes (fetch, score, resume, apply, notify, report) ■ ■ ■■■■ orchestrator.py # LangGraph StateGraph
compilation ■ ■ ■■■■ state.py # AgentState TypedDict schema ■ ■■■■ config/ # Central Configuration & Source
Settings ■ ■ ■■■■ job_sources.yml # YAML source enable/disable & priority config ■ ■ ■■■■ settings.py #
Environment variables & candidate profile ■ ■■■■ connectors/ # 16 Active Job Ingestion Connectors ■ ■ ■■■■
base.py # BaseConnector abstract interface ■ ■ ■■■■ manager.py # SearchManager concurrent executor ■ ■ ■■■■
registry.py # Connector discovery decorator ■ ■ ■■■■ greenhouse_connector.py # Greenhouse ATS API ■ ■ ■■■■
lever_connector.py # Lever ATS API ■ ■ ■■■■ ashby_connector.py # Ashby ATS API ■ ■ ■■■■ workday_connector.py
# Workday CXS Portal ■ ■ ■■■■ smartrecruiters_connector.py # SmartRecruiters API ■ ■ ■■■■ jobicity_connector.py
# Jobicity Remote API ■ ■ ■■■■ remotive_connector.py # Remotive Software API ■ ■ ■■■■ himalayas_connector.py #
Himalayas Remote API ■ ■ ■■■■ adzuna_connector.py # Adzuna Jobs API ■ ■ ■■■■ agent_reach_connector.py # Agent
Reach LinkedIn scraper ■ ■ ■■■■ linkedin_posts_connector.py # LinkedIn Hiring Posts parser ■ ■ ■■■■
ycombinator_connector.py # HackerNews / YC Algolia API ■ ■ ■■■■ wellfound_connector.py # Wellfound AI
Startups ■ ■ ■■■■ arbeitnow_connector.py # Arbeitnow Tech API ■ ■ ■■■■ weworkremotely_connector.py #
WeWorkRemotely RSS ■ ■■■■ db/ # Database Storage Layer ■ ■ ■■■■ models.py # SQLAlchemy ORM models (Job,
Application, Resume) ■ ■ ■■■■ session.py # SQLite engine (jobs.db) ■ ■■■■ prompts/ # LLM System Personas &
Prompts ■ ■ ■■■■ email_prompt.py # Cold outreach email prompt ■ ■ ■■■■ resume_prompt.py # Dynamic ATS resume
tailoring prompt ■ ■ ■■■■ scoring_prompt.py # Strict 0-2 yrs technical scoring prompt ■ ■■■■ schemas/ # Data
Transfer Objects ■ ■ ■■■■ application.py # Application status schema ■ ■ ■■■■ job_data.py # Unified JobData
dataclass ■ ■■■■ services/ # Core Business Logic ■ ■ ■■■■ application_policy.py # Source-Aware 3-Mode Policy
Manager ■ ■ ■■■■ apply_service.py # Application routing & submission ■ ■ ■■■■ browser_apply_service.py #
Playwright Chromium persistent browser engine ■ ■ ■■■■ email_service.py # Gmail API OAuth2 sender ■ ■ ■■■■
mock_scoring_service.py # Heuristic pre-filter engine ■ ■ ■■■■ report_service.py # openpyxl multi-tab Excel
generator ■ ■ ■■■■ resume_service.py # Dynamic ATS 1-page PDF builder ■ ■ ■■■■ scheduler_service.py #
APScheduler 24/7 6-hour cron service ■ ■ ■■■■ scoring_service.py # Gemini AI semantic scoring ■ ■ ■■■■
whatsapp_service.py # CallMeBot & Meta WhatsApp alerts ■ ■■■■ utils/ # Validators & Helper Utilities ■ ■■■■
email_validator.py # Explicit HR email parser & DNS validator ■ ■■■■ experience_analyzer.py # Dedicated 0-2
yrs ExperienceAnalyzer ■ ■■■■ experience_filter.py # Senior title & experience hard guard ■ ■■■■
gemini_client.py # Resilient Gemini client with retry ■■■■ data/ # Persistent Storage & Browser Profiles ■
■■■■ browser_profile/ # Playwright persistent Chromium session & cookies ■ ■■■■ jobs.db # SQLite primary
database ■■■■ docs/ # Architecture Documentation & PDF Blueprints ■ ■■■■
AI_Job_Agent_Architecture_and_Workflow.pdf ■■■■ reports/ # Daily Multi-Tab Excel Reports ■■■■ resumes/ #
Master & Tailored ATS PDF Resumes ■■■■ scripts/ # Operational Scripts & Tools ■ ■■■■ run_agent.py # CLI runner
for single pipeline execution ■ ■■■■ test_browser_apply.py # Live visible browser auto-apply tester ■ ■■■■
setup_gmail_token.py # Gmail OAuth2 token generator ■ ■■■■ setup_whatsapp.py # WhatsApp CallMeBot diagnostic
tool ■ ■■■■ generate_architecture_pdf.py # Architecture PDF builder ■■■■ main.py # FastAPI 24/7 Server &
Background Scheduler Entrypoint ■■■■ requirements.txt # Production Python dependencies ■■■■ .env #
Configuration & API Credentials

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